METRIC

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25 April 2022

SUPERSEDING
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6 December 2011

PERFORMANCE SPECIFICATION

LUBRICATING OIL, INSTRUMENT, AIRCRAFT, LOW VOLATILITY

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This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one grade of aircraft instrument lubricating oil only (see 6.1). The lubricating oil is identified by military symbol OAI and NATO Code No. O-147 (see 6.4).

1.2 Part or identifying number (PIN). The PIN to be used for lubricating oil acquired to this specification is created as follows:

M6085 - X

Container Size
1 = 1.5 oz.
2 = 4 oz.
3 = 8 oz.
4 = 1 quart
5 = 5 gallon

2 APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

QQ-S-698 - Steel, Sheet and Strip, Low Carbon

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-680 - Degreasing Solvent

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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ASTM INTERNATIONAL

ASTM A29/A29M	-	Stand Specification for General Requirements for Steel Bars, Carbon and Alloy, Hot-Wrought
ASTM B36/B36M	-	Standard Specification for Brass Plate, Sheet, Strip, And Rolled Bar
ASTM D91	-	Standard Test Method for Precipitation Number of Lubricating Oils
ASTM D92	-	Standard Test Method for Flash and Fire Points by Cleveland Open Cup Tester
ASTM D97	-	Standard Test Method for Pour Point of Petroleum Products
ASTM D445	-	Standard Test Method for Kinematic Viscosity of Transparent and Opaque Liquids (and Calculation of Dynamic Viscosity)
ASTM D972	-	Standard Test Method for Evaporation Loss of Lubricating Greases and Oils
ASTM D974	-	Standard Test Method for Acid and Base Number by Color-Indicator Titration
ASTM D1500	-	Standard Test Method for ASTM Color of Petroleum Products (ASTM Color Scale)
ASTM D1748	-	Standard Test Method for Rust Protection by Metal Preservatives in the Humidity Cabinet
ASTM D4057	-	Standard Practice for Manual Sampling of Petroleum and Petroleum Products
ASTM D4636	-	Standard Test Method for Corrosiveness and Oxidation Stability of Hydraulic Oils, Aircraft Turbine Engine Lubricants, and Other Highly Refined Oils
ASTM D5949	-	Standard Test Method for Pour Point of Petroleum Products (Automatic Pressure Pulsing Method)
ASTM D5950	-	Standard Test Method for Pour Point of Petroleum Products (Automatic Tilt Method)

(Copies of these documents are available from https://www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. The instrument oil furnished under this specification shall be products authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Materials. Aircraft instrument lubricating oil (low volatility) shall consist substantially of synthetic base oil and additive materials to impart oxidation stability and corrosion-protective properties. Certain esters of the dibasic organic acids containing six to ten carbon atoms are known to be suitable for use but are not mandatory. The right is reserved to subject oils of new or unusual composition to such additional tests as are considered necessary to ensure serviceability of the material. Viscosity-temperature coefficient (VI) improvers or pour point depressants shall not be used.

3.3 Properties. Products shall conform to the requirements as specified in Table I and 3.4 through 3.6 when tested as specified in Section 4.

TABLE I. Properties of the finished oil.

Characteristic	Value
Viscosity in centistokes (mm^2/s)	
at 54°C (min)	8
at -54°C (max)	12,000
Pour point (max)	-57°C
Flash point (min)	185°C
Precipitation number (max)	0
Acid number	report

3.4 Performance.

3.4.1 Corrosion and oxidation stability. When tested as specified in section 4, the change in weight of cadmium-plated steel, copper, steel aluminum alloy, and magnesium alloy, when subjected to the action of the oil for 168 hours at 121°C ± 1°C, shall be not greater than 0.2 mg per square cm of surface for each strip. There shall be no pitting, etching, or visible corrosion on the surface of any of the metals when viewed under magnification of 20 diameters. Slight discoloration of the surface of the copper will be permitted, but dark brown, gray or black stain shall be cause for rejection.

3.4.2 Resistance to oxidation. When tested as specified in section 4, the oil shall not have changed more than ±5 percent from the original viscosity in centistokes at 54.4°C after the oxidation-corrosion test (see 4.4.2). The neutralization number increase shall be not greater than 0.5 after oxidation. There shall be no evidence of separation of insoluble materials or gumming of the oil.

3.4.3 Low temperature stability. When tested as specified in section 4, there shall be no gelling or separation of solid or liquid phases in the oil after storage at a temperature at or below -54°C for 72 hours. The presence of a dense cloud, which does not “settle out”, shall not be the cause for rejection.

3.4.4 Evaporation loss. When tested as specified in section 4, there shall be not more than 1.80 percent by weight evaporation loss after the oil has been subjected to the test conditions, at 120°C ± 1°C.

3.4.5 Corrosivity. When subjected to the test specified in section 4, the test areas of the three discs shall show no evidence of corrosion, pitting, or other attack. The discs may show no more than three spots within the area covered by the clip.

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3.5 Color and appearance. When tested as specified in section 4, the oil shall be clear, transparent and uniform in appearance, and not darker than No. 5 Union Colorimeter of ASTM D1500.

3.6 Protection of panels. When tested as specified in section 4, not more than one panel out of five panels shall fail after being covered with a film of lubricating oil for 100 hours. If more than one panel fails, the product shall be retested by repeating the same test with an additional ten panels. Not more than four panels shall fail out of the total 15 panels, (the five panels on the original test plus the ten panels on retest).

3.7 Workmanship. The oil shall be entirely homogeneous and free from lumps of undissolved additive, water, dirt, lint, or sediment. Prior to final packaging, the oil shall be filtered through a 0.45-micrometer pore-size filter.

3.8 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs."

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 Qualification inspection. Qualification inspection shall consist of a review for approval of the submitted manufacturers test report and subjecting the qualification samples (see 4.2.1) to examination and testing for all the requirements of this specification.

4.2.1 Qualification samples. The qualification sample(s) shall consist of 1 gallon finished oil.

4.2.2 Qualification tests. Qualification sample(s) shall be subjected to all tests specified in 4.4, method of inspection.

4.2.3 Retention of qualification. In order to retain qualification of approval for listing on the QPL, the manufacturer shall verify, by certification (DD Form 1718) to the Qualifying Activity that the manufacturer's product complies with the requirements of this specification. The manufacturer shall also provide a recent Certificate of Analysis (COA) to include all conformance inspection requirements (see 4.3). The time of periodic verification by certification shall be in two-year intervals from the date of original qualification. The Government reserves the right to re-examine the qualified product whenever deemed necessary to determine the product continues to meet any or all of the specification requirements.

4.2.4 Requalification. Requalification shall be required when any change is made in the source of manufacture, purity, or composition of the fluid's base stock or additives. A minor change in the fluid formulation may be made without requalification testing, but only after notification to, and approval by the qualifying activity (see 6.3). The qualifying activity may, at its discretion, waive complete requalification or may require only partial requalification testing to determine the significance and acceptability of the proposed formulation change.

4.3 Conformance inspection. Conformance samples shall be labeled completely with information identifying the purposes of the sample, name of product, specification number, lot and batch number (packaged or bulk - see 6.7), date of sampling, and contract number. Quality conformance tests shall be as specified in Table II.

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4.3.1 Sampling. Sampling shall be performed in accordance with ASTM D4057.

TABLE II. Quality conformance tests.

Inspection	Paragraph Requirement	Test
Corrosion and oxidation stability	3.4.1, 3.4.2	4.4.2
Low temperature stability	3.4.3	4.4.2
Viscosity	3.3	4.4.2
Pour point	3.3	4.4.2
Flash point	3.3	4.4.2
Precipitation number	3.3	4.4.2
Evaporation loss	3.4.4	4.4.2
Examination of filled containers	5.1	4.4.3
Color	3.5	4.4.2

4.4 Method of inspection.

4.4.1 Inspection. Inspection shall be in accordance with 4.4.3 of this specification.

4.4.2 Test Methods. The oil properties shall be determined in accordance with the applicable methods specified in Table III and 4.4.1 through 4.4.3. Physical and chemical values specified in section 3 apply to the arithmetic average of the determinations made on the samples for those values that fall within any stated repeatability or reproducibility limits of the applicable test method.

TABLE III. Test methods for oil properties.

CHARACTERISTIC	ASTM TEST METHODS
Corrosion and oxidation stability	D4636 1/
Pour point	D97, D5949, or D5950
Flash point (open cup method)	D92
Precipitation number	D91
Viscosity (kinematic)	D445
Color	D1500
Evaporation loss of lubricating greases and oils	D972
Acid number	D974

Note:

1/ Test shall be run for 168 hours at 121°C. Use hexane, heptane, and/or acetone to clean coupons. Use Alternate Procedure 2.

4.4.3 Examination of filled container. Each sample of the filled unit container and the shipping container shall be examined for evidence of leakage and net content, and examined for construction defects in the container and closure. Any container in the sample that is under the required fill, or has one or more defects shall be rejected. Rejected lots may be resubmitted for

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acceptance inspection provided that the contractor has removed or repaired all nonconforming containers.

4.4.4 Protection test.

4.4.4.1 Preparation of panels. The following procedures shall be conducted:

a. Cut five panels from steel that conforms to QQ-S-698.

b. Size the panels and locate the holes as shown in Figure A1.6, Panel and Suspension Hooks, of ASTM D1748. Remove all burrs, sharp edges, and corners, including the edges of the holes.

c. Highly polish the panel with No. 3/0 emery paper prior to use. Next wash and clean the panel in a solvent conforming to MIL-PRF-680, Type 1. Drain thoroughly and then agitate in boiling 95-percent or higher methanol.

d. After cleansing, cool the panels in a desiccator. Care must be taken during cleaning and preparation to ensure that the surfaces are not contaminated by fingerprints. Handle the panels only with tongs during the cleaning operation and only with hooks during and after dipping.

4.4.4.2 Procedure. These steps shall be conducted:

a. Dip five panels in a suitable sample of lubricating oil which is maintained at 25°C ± 2°C, remove and allow to drain for 2 hours at that temperature from glass, Monel, or stainless steel supports.

b. At the end of this period, suspend the panels in a humidity cabinet conforming to ASTM D1748 for a period of 100 hours, and in such a manner that the drippage from the supports will not fall onto the panels. Maintain the humidity within the cabinet at 100 percent relative humidity and at a dry bulb temperature of 49°C ± 2°C for a 100-hour exposure period.

c. Remove the panels from the cabinet, clean them with naphtha and examine them. Panels have failed the protection test if at the end of the test period one of the following conditions exists in the significant areas of the panels, as defined by ASTM D1748, Figure A1.7, Significant Area of Test Panel, considering both sides of the panel:

(1) A corroded area of 2-mm diameter or larger

(2) Two or more spots of corrosion between 1 and 2 mm maximum diameter.

d. If more than one panel from the five fails the protection test as defined above, a retest will be permitted. Retests shall consist of repeating the protection test using ten additional panels. Reject the lubricating oil if more than four panels fail the test (adding failures of both test and retest).

4.4.5 Stability at low temperature. A 100-ml sample of the oil shall be maintained at or below -54°C for 72 hours in a stoppered flask. At the end of that time there shall be no visual evidence of gelling, separation or crystallization of the oil. Presence of a dense cloud, which does not “settle out” shall not be cause for rejection.

4.4.6 Corrosivity test.

4.4.6.1 Preparation of panels (also see 4.4.4.1).

4.4.6.2 Preparation of steel discs. The following steps shall be performed:

a. Cut three discs 12.5 mm thick from a 25 mm diameter bar stock that conforms to ASTM A29/A29M, composition E52100.

b. Heat-treat the discs to a hardness of Rockwell C-62. (Rollers that may be obtained from roller bearings with similar chemical, physical and dimensional properties to the above bar stock after heat treatment can be substituted for the formation of the discs). In this case, no additional heat treatment is necessary.

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c. Slowly surface grind on one side to a finish of less than 20 micro-inches rms. If coolant is used during grinding, slush the discs in absolute methanol. Next, abrade the discs on the surface-ground side with successive applications of emery polishing paper grades 1/0, 2/0, 3/0, and finally 4/0. Papers incorporating iron oxide as the polishing medium and wet-dry type papers shall not be used.

d. Wipe the discs clean with sterile absorbent gauze and examine under a 10X magnification for any signs of corrosion or other defects. Do not use defective specimens. Store the discs in a desiccator containing silica gel until ready for use.

4.4.6.3 Preparation of brass clips. The following steps shall be performed:

a. Fabricate clips from commercial 0.6 mm brass sheet conforming to ASTM B36/B36M spring temper. The size and shape of the clips are illustrated in Figure 1.

b. Immerse the clips for 20 seconds in an etching solution comprised of:

450 mL water
300 mL conc. sulfuric acid
225 mL conc. nitric acid
8 mL conc. hydrochloric acid

c. After etching, wash the clips in cold running tap water followed by distilled water. Dry with acetone and store the clips in a desiccator containing silica gel as desiccant until ready for use.

4.4.6.4 Procedure. The following procedures shall be performed:

a. Coat the three discs with the test oil by dipping and stirring the rod in the test oil and allowing the oil from the rod to drop onto the polished side of the discs.

b. Clamp the brass clips over the coated discs and the assemblies and place them in a test chamber, which is maintained at 27°C ±1°C and 50 percent relative humidity, for a period of 10 days. For a 50 percent relative humidity, use a sulfuric acid solution with a density of 1.338 g/mL at 20°C.

c. After exposure, remove the assemblies and outline the brass clips on the discs using the clips as templates. Then remove the clips and wipe the test oil from the discs.

d. Examine the discs under 10X magnification for signs of corrosion, pitting or other detrimental effects. If the test results are questionable, repeat the entire test using new specimens, except extend the period to 20 days. In this case, examine the discs as in the previous manner period to 20 days. In this case, examine the discs as in the previous manner.

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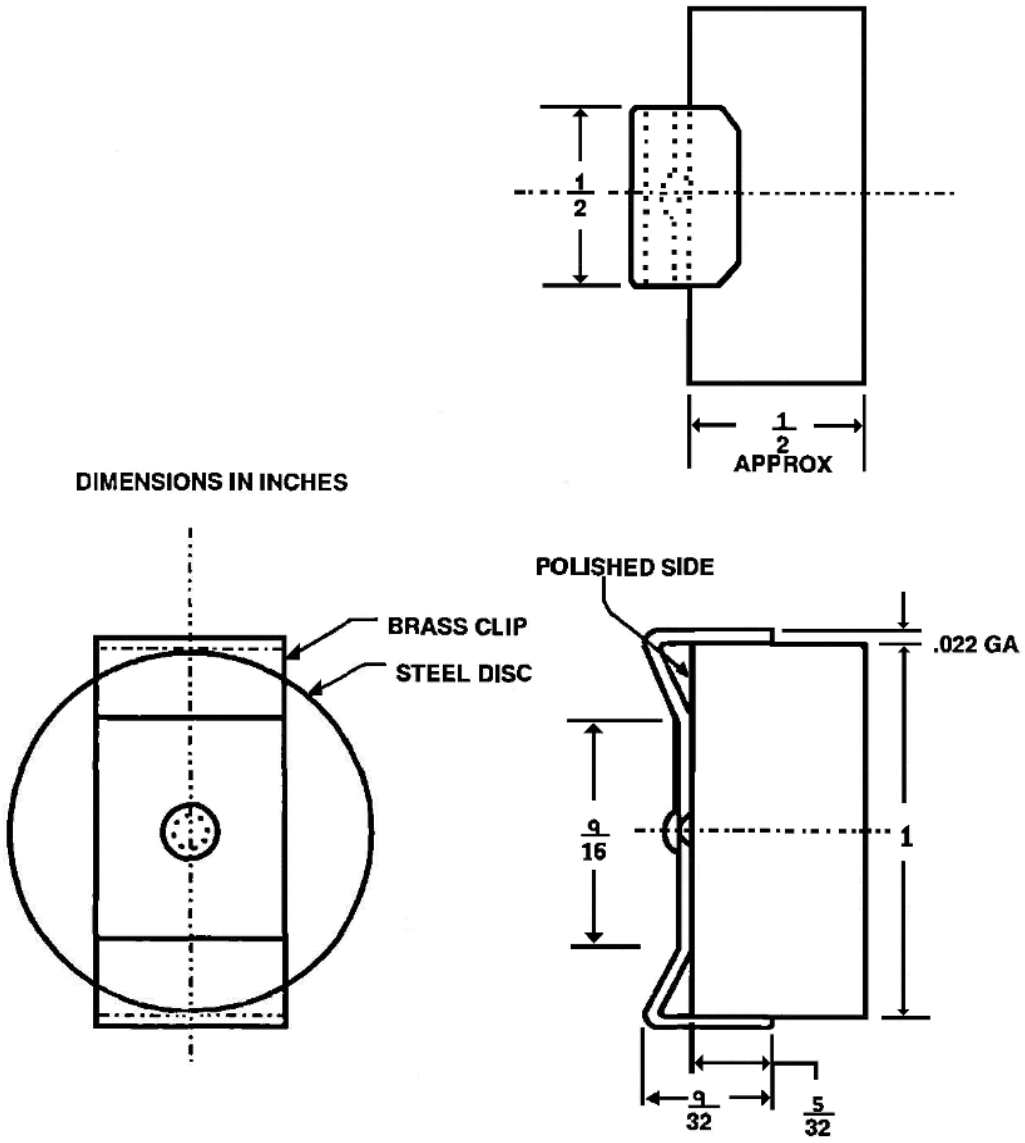


Figure 1. Typical test specimens for corrosivity test

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This oil is intended for use in military aircraft instruments, electronic equipment or where low evaporation oil is required for both high and low temperature application, and where oxidation and corrosion resistance is desirable.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. If required, the specific issue of the individual documents referenced (see 2.1 and 2.2).

c. Type and size of containers (see 5.1).

d. Quantity.

e. Selection of applicable levels of packaging and packing with requirements in detail (see 5.1).

6.2.1 Purchase unit. The instrument oil covered by this specification should be purchased by volume.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set of award of contract, qualified for inclusion in Qualified Products List QPL No. 6085 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from AFPET/PTPS, 2430 C Street, Building 70, Area B, Wright-Patterson AFB OH 45433, AFPET.PTPS@us.af.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.3.1 Qualification information. It is understood that the material furnished under this specification subsequent to final approval will be of the same composition and will be equal to products upon which approval was originally granted. In the event that the oil furnished under contract is found to deviate from the composition of the approval product, or that the product fails to perform satisfactorily, approval of such products will be subjected to immediate withdrawal from the QPL at the discretion of the approving activity.

6.3.2 Data to accompany qualification samples. Samples will be accompanied by a safety data sheet (SDS) and a test report from the manufacturer or a commercial laboratory containing complete information about the source, type of base stock and additive materials used; the formulation and composition of the finished fluid; and laboratory data showing quantitative results of all the tests required by this specification, except storage stability. Separate qualification inspection

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will be required for each base stock used. The samples will be plainly identified by securely attached durable tags or labels marked with the following information:

Sample for qualification inspection
LUBRICATING OIL; AIRCRAFT INSTRUMENT, LOW VOLATILITY
Specification *MIL-PRF-6085*
Name of manufacturer
Product code number
Date of manufacture

6.4 International standardization agreement implementation. This specification implements certain provisions of AFIC Air Standard 4024 *Interchangeability Chart of Standardised Aviation Fuels, Lubricants, and Associated Products* and NATO STANAG 1135 *Interchangeability of Fuels, Lubricants and Associated Products Used by the Armed Forces of the North Atlantic Treaty Nations*. When amendment, revision, or cancellation of this specification is proposed, the preparing activity must coordinate the action with the U.S. National Point of Contact for the international standardization agreement, as identified in the ASSIST database at https://assist.dla.mil.

6.5 Shelf-life. This specification covers items where the assignment of a Federal shelf-life code is a consideration. Specific shelf-life requirements should be specified in the contract or purchase order, and should include, as a minimum, shelf-life code, shelf-life package markings IAW MIL-STD-129 or FED-STD-123, preparation of a materiel quality storage standard for type II (extendible) shelf-life items, and a minimum of 85 percent shelf-life remaining at time of receipt by the Government. These and other requirements, if necessary, are in DoDM 4140.27, Volume 1, *Shelf-Life Management Program: Program Administration*; and Volume 2, *Shelf-Life Management Program: Materiel Quality Control Storage Standards*. The shelf-life codes are in the Federal Logistics Information System Total Item Record. Additive information for shelf-life management may be obtained from DoDM 4140.27 Volumes 1 and 2, or the designated shelf-life Points of Contact (POC). The POC should be contacted in the following order: (1) the Inventory Control Points that manage the item and (2) the DoD Service and Agency administrators for the DoD Shelf-Life Program. Appropriate POCs for the DoD Shelf-Life Program can be contacted through the DoD Shelf-Life Management website: https://www.shelflife.dla.mil/.

6.6 Subject term (keyword) listing.

Acid number
Flash point
Pour point
Synthetic lubricant

6.7 Definitions.

6.7.1 Bulk lot. A bulk lot (batch) is an indefinite quantity of a homogeneous material mixture offered for acceptance in a single, isolated container or manufactured in a single-plant run (not exceeding 24 hours) through the same processing equipment, with no change in ingredient material.

6.7.2 Packaged lot. A packaged lot is an indefinite number of unit containers of identical size and type, offered for acceptance and filled with a homogeneous material mixture from one isolated container or filled with a material mixture manufactured in a single-plant run (not exceeding 24 hours) through the same processing equipment, with no change in ingredient material.

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes. The changes are due to Acquisition Reform initiatives requiring Government specifications to be performance-based.

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Preparing activity:
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